



### PROPOSED SOLUTION & CORE CAPABILITIES:

- **Continuous Post-Complaint Care:** Proactively monitors atrocity victims (SC/ST PoA Act 1989 / NHAA 14566) across legal trial & rehabilitation phases with automated check-ins.
- **Omnichannel Multimodal Ingestion:** Hooks into NHAA 14566 SIP lines, Citizen Web portal, Chatbot API, SMS & Mobile App with DPDP-compliant consent.
- **Dynamic Distress Scoring (DDS 0–100):** Continuous multi-vector metric  $DDS = 0.25E + 0.20V + 0.15B + 0.15H + 0.10C + 0.15T$  combining emotion valence, vocal tremor, behavior & legal delays.
- **72-Hour Longitudinal Crisis Forecasting:** Computes distress trend velocity ( $dDDS/dt$ ) over 7-day rolling window to forecast crisis 72 hours prior to acute escalation.
- **3-Tier Risk Bands & Action Matrix:** 0–30: Self-help grounding audio | 31–60: Tele-counselling intervention | 61–100: Real-time emergency alert to DM / SP dashboard & crisis dispatch.
- **Explainable AI (XAI) & Human Gate:** SHAP/LIME point attributions (+24 threat cue, +18 vocal tremor, +15 case delay) ensure auditable decision transparency.

### Victim Experience & Multichannel Hub (14566, Web, Chatbot & App)

- **Omnichannel Multi-Access:** Direct toll-free 14566 calling, Citizen Web portal, Tanglish Chatbot & Mobile App SDK.
- **Tanglish & Multilingual AI:** Native support for Tanglish (Tamil-English), Tamil, Hindi & English with voice dialogue.
- **Calm Space & Silent Sanctuary:** Interactive emotion wheel, Box Breathing (4-4-4-4) & non-verbal Silent Mode tracking.

### 3-Tier Command HUD: District (DM/SP) -> State Nodal -> National (MoSJE)

- **District Level (DM, SP, DLSA):** Case-level telemetry, XAI factor cards, local crisis team & Sec 15A protection.
- **State Level (State Nodal Officer):** Inter-district resource allocation, statewide distress heatmap & Rule 12(4) tracking.
- **National Level (MoSJE Central):** Centrally sponsored scheme oversight, national vulnerability trends & policy analytics.

#### 72-Hour Crisis Early Warning

Distress velocity ( $dDDS/dt$ ) predicting crises 72 hours prior.

#### Tanglish & Low-Bandwidth Mode

Full offline grounding audio, low-data layout & SMS fallback.

#### 3-Tier Multi-Agency Action Matrix

Self-help -> Tele-counselling -> DM/SP witness protection.

## Idea / System Architecture Flow:

### 1. Omnichannel Ingest

14566 SIP PRI lines, WebRTC mic, Chatbot API & Mobile App SDK (8/16kHz PCM)

### 2. Multimodal Signal Extraction

Vocal acoustic prosody (pitch tremor Hz, jitter %, pause duration) + BERT fear valence

### 3. Dynamic Distress Scoring (0–100)

Continuous formula:  $DDS = 0.25E + 0.20V + 0.15B + 0.15H + 0.10C + 0.15T$

### 4. 72h Crisis Forecast & Explainable AI

Time-series trend velocity (dDDS/dt) predicts crisis 72h prior + SHAP point attributions

### 5. Automated 3-Tier Risk Action Dispatch

0–30: Self-help grounding audio | 31–60: Tele-counselling | 61–100: Emergency DM/SP alert

## Technical Stack & Security:

- **Speech & Emotion DSP:** Python (NumPy, SciPy, Librosa) 20ms frames, vocal tremor analysis.
- **AI / NLP & Predictive Engine:** Hugging Face Transformers BERT, velocity forecasting (dDDS/dt), XAI.
- **Telemetry & Secure Ledger:** FastAPI ASGI server, PostgreSQL, Redis cache, AES-256 GCM vault.

## Workflow Comparison Matrix:

| WORKFLOW FEATURE        | CONVENTIONAL WORKFLOW            | KEFI AI PLATFORM                                          |
|-------------------------|----------------------------------|-----------------------------------------------------------|
| Voice Triage            | Manual assessment & referral     | ✓ <b>Proactive Care</b><br>Scheduled check-ins            |
| Multilingual & Channels | Varies by channel & availability | ✓ <b>22+ Languages</b><br>Tanglish / Web / 14566          |
| Triage Framework        | No standardized composite index  | ✓ <b>Mathematical DDS</b><br>0–100 multi-signal metric    |
| Emergency Escalation    | Manual emergency workflow        | ✓ <b>72h Early Warning</b><br>Predictive velocity dDDS/dt |

### 20ms Frame Acoustic DSP

Linear PCM stream ingestion, pitch & vocal micro-tremor extraction.

### Mathematical DDS Architecture

$DDS = 0.25E + 0.20V + 0.15B + 0.15H + 0.10C + 0.15T$   
multi-signal metric.

### AI Statutory Recommendation Engine

NLP triggers 4 specific actions: Medical care, Sec 15A Relocation, DLSA Legal aid & Rule 12(4) DBT.

## ANALYSIS OF THE FEASIBILITY:

- **Technical Feasibility:** Prototype benchmarked with initial acoustic DSP & NLP inference latency below 250ms.
- **Operational Feasibility:** Seamless integration with MoSJE NHAA 14566 PRI lines, Web portal & App.
- **Legal & Ethical Feasibility:** Aligned with SC/ST PoA Act 1989 (Sec 15A & Rule 12(4)) and DPDP Act 2023.

## VIABILITY & SCALABILITY:

- **Cost Viability:** Projected prototype operating cost: <₹0.15 per check-in; zero costly hardware required.
- **Reliability & Scalability:** High-availability cloud architecture with offline caching, Low Internet Mode & fallback.

## PROBLEM VS SOLUTION

### ✖ PROBLEMS

- **Subjective Triage:** Check-ins are irregular, subjective, and absent during legal delays.
- **Delayed Detection:** Distress escalation goes unnoticed until acute harm occurs.
- **Witness Threats:** Coercion goes undetected, causing witness hostility in court.
- **Disconnected Logs:** Paper registers leave DM & SP unaware of localized crisis trends.
- **High Workload:** Manual welfare follow-up is impossible across thousands of cases.

### ✔ SOLUTION

- **Objective Scoring:** Automated continuous monitoring via mathematical DDS metric (0–100).
- **72h Early Warning:** Longitudinal velocity (dDDS/dt) alerts counsellors 72h prior to crisis.
- **Threat Screening:** NLP flags coercion keywords ('dhamki', 'withdraw') for Sec 15A protection.
- **Unified HUD:** Live map alerts DM, SP & MoSJE to localized distress velocity spikes.
- **70%+ Overhead Cut:** Automated check-ins and AI triage free officers for high-risk care.

### High-Availability Architecture

Tested prototype; sub-250ms median API latency; automated failover.

### DPDP Act 2023 & Rules 2025

SHA-256 tamper-evident audit ledger & AES-256 GCM encrypted audio vault.

### Fail-Safe & Low-Confidence

Deterministic P1 safety override + low-confidence human verification fallback.

## Key Impacts on Victims & Institutions:

01. **72-Hour Suicide & Crisis Prevention:** Critical distress detected 72h prior, enabling timely tele-counselling & Sec 15A protection.
02. **Single-Window Trauma Continuity:** Victim recounts ordeal once; structured statutory dossier routed to all departments.
03. **Witness Protection & Justice Trust:** Real-time threat detection triggers Sec 15A protection, reducing hostile witness rates.
04. **Targeted Statutory Redress:** AI NLP precisely recommends Medical care, Sec 15A Relocation, DLSA Legal aid & Rule 12(4) DBT relief.
05. **70%+ Administrative Overhead Cut:** Automated check-ins and smart triage free officers for complex crisis response.

## PS Expected Outcomes Compliance Matrix:

| PS EXPECTED OUTCOME        | KEFI AI IMPLEMENTATION                                                                  | STATUS      |
|----------------------------|-----------------------------------------------------------------------------------------|-------------|
| 1. Early Identification    | Automated scheduled check-ins via IVRS 14566, Chatbot, SMS & Web portal                 | ✓ Addressed |
| 2. Counselling & Rehab     | Mathematical formula DDS (0–100) combining 6 trauma signals & delay factors             | ✓ Addressed |
| 3. Victim-Centric Redress  | 72-hour crisis velocity forecasting (dDDS/dt) predicting acute trauma before escalation | ✓ Addressed |
| 4. Resource Allocation     | 3-Tier risk dispatch: DM/SP emergency alert, tele-counselling & self-help grounding     | ✓ Addressed |
| 5. Enhanced Responsiveness | DPDP Act 2023 compliance, AES-256 encryption, OTP consent & audit trail                 | ✓ Addressed |

### Outcome 1: Tanglish Chatbot

### Outcome 2: Emotion Wheel

### Outcome 3: Calm Sanctuary

### Outcome 4: 7-Day DDS Trend

### Outcome 5: Officer HUD

Statutory Acts & Policy References:

- **Scheduled Castes & Scheduled Tribes (PoA) Act, 1989:** Section 15A (Rights of victims & witnesses, including protection) & Section 18A (FIR registration provisions).
- **SC & ST (PoA) Rules, 1995:** Rule 12(4) — Relief and Rehabilitation framework, including prescribed relief provisions and seven-day service requirement.
- **Digital Personal Data Protection:** DPDP Act, 2023 and DPDP Rules, 2025 for data fiduciary duties & consent.
- **National Helpline Integration:** MoSJE National Helpline Against Atrocities (NHAA 14566) continuous linkage.

Scientific Research & AI Citations:

- **Acoustic Emotion Prosody:** Scherer, K. R. (2003). Vocal communication of emotion. Speech Communication.
- **Voice Tremor & Distress:** Mendoza, J. et al. (2019). Acoustic biomarkers of PTSD in call audio. IEEE TAC.
- **Explainable AI (XAI):** Lundberg, S. M. & Lee, S.-I. (2017). Unified SHAP approach for interpretable ML. NeurIPS.
- **Clinical Trauma Criteria:** American Psychological Association (APA) - DSM-5 Diagnostic Criteria for PTSD.

TEAM DEKO (TEAM ID: 141444) — UNIVERSITY COLLEGE OF ENGINEERING, PANRUTI

| TEAM MEMBER            | STREAM | BRANCH | YEAR    | PROJECT ROLE & KEY RESPONSIBILITY                                           |
|------------------------|--------|--------|---------|-----------------------------------------------------------------------------|
| BALAJI P (Team Leader) | B.E    | CSE    | IV Year | System Architect, Dynamic Distress Scoring (DDS) & Predictive Engine Lead   |
| NAVANEETHAM V          | B.E    | CSE    | IV Year | Multilingual Tanglish NLP, Bhashini ASR & Emotion Wheel Architecture Lead   |
| MALINI V               | B.E    | CSE    | IV Year | Full-Stack Web Portal, Calm Space & Silent Sanctuary Lead                   |
| DHAVAN R G             | B.E    | CSE    | IV Year | DM/SP Dashboard, Automated Alert Matrix & Incident Dispatch Lead            |
| DEEBIKA S              | B.E    | CSE    | II Year | Cloud Security, AES-256 GCM Audio Vault & DPDP Rules 2025 Compliance Lead   |
| DHARA R                | B.E    | CSE    | II Year | Rule 12(4) Relief Tracking, IVRS 14566 Integration & Citizen Portal UI Lead |